

WHITEBRIDGE HIGH SCHOOL

2012

Student Name:

HIGHER SCHOOL CERTIFICATE
ASSESSMENT 4

Mathematics Extension 2

General Instructions

- Reading Time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen.
Black pen is preferred
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- Show all necessary working in Questions 11 to 16

Total marks – 100

Section I

10 marks

- Attempt Questions 1 to 10
- Allow about 15 minutes for this section
- Use the multiple choice answer sheet

Section II

60 marks

- Attempt Questions 11 – 14
- Allow about 2 hours 45 minutes for this section

This paper MUST NOT be removed from the examination room

Section I**10 Marks****Attempt Questions 1 to 10****Allow about 15 minutes for this section**

Select the alternative A, B, C or D that best answers the question and indicate your choice by colouring in the bubble that corresponds to your answer.

		Marks
1	Which of the following is an expression for $\int \frac{\cos^3 x + \sin^3 x}{\cos x + \sin x} dx$?	1
(A) $x + \frac{1}{4} \cos 2x + c$		
(B) $x - \frac{1}{4} \cos 2x + c$		
(C) $x + \frac{1}{2} \sin 2x + c$		
(D) $x - \frac{1}{2} \sin 2x + c$		
2	A particle moving in a straight line is performing Simple Harmonic Motion. At time t seconds it has displacement x metres from a fixed point O on the line and velocity $v \text{ ms}^{-1}$ given by $v^2 = 9(5 + 4x - x^2)$. Where is the centre of motion?	1
(A) $x = -1$		
(B) $x = 0$		
(C) $x = 2$		
(D) $x = 5$		
3	If $x = \theta - \sin \theta$ and $y = 1 - \cos \theta$, which of the following is an expression for $\frac{dy}{dx}$?	1
(A) $\cot^2 \frac{\theta}{2}$		
(B) $\cot \frac{\theta}{2}$		
(C) $\tan \frac{\theta}{2}$		
(D) $\tan^2 \frac{\theta}{2}$		

Marks

- 4 In the Argand Diagram the locus of the point P representing the complex number z such that $|z - 1 + i| = 4$ is a circle. What are the centre and radius of this circle?

1

- (A) centre $(-1, 1)$ and radius 4
(B) centre $(-1, 1)$ and radius 2
(C) centre $(1, -1)$ and radius 4
(D) centre $(1, -1)$ and radius 2

- 5 The equation $x^3 + 2x + 1 = 0$ has roots α , β and γ .

1

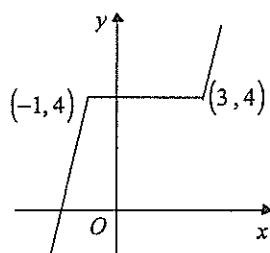
Which of the following equations has roots 2α , 2β and 2γ ?

- (A) $x^3 + 8x + 8 = 0$
(B) $x^3 + 16x + 8 = 0$
(C) $2x^3 + 4x + 2 = 0$
(D) $8x^3 + 4x + 1 = 0$

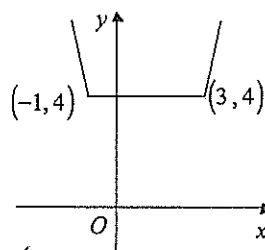
- 6 Which of the following is the graph of $y = |x + 1| + |x - 3|$?

1

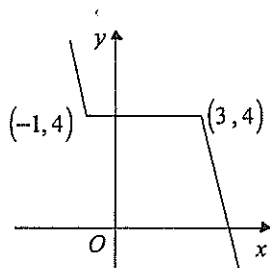
(A)



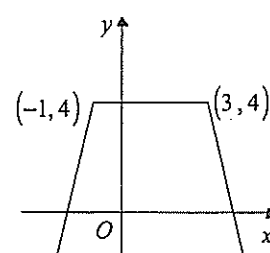
(B)



(C)



(D)



Marks

- 7** The normal at the point $P(cp, \frac{c}{p})$ on the rectangular hyperbola $xy = c^2$ has equation $p^3x - py = cp^4 - c$. This normal cuts the hyperbola at a second point $Q(cq, \frac{c}{q})$. What is the relationship between q and p ?

1

- (A) $p^4q = -1$
(B) $p^3q = -1$
(C) $p^2q = -1$
(D) $pq = -1$

- 8** Which of the following is an expression for $\int x e^{\frac{x}{2}} dx$?

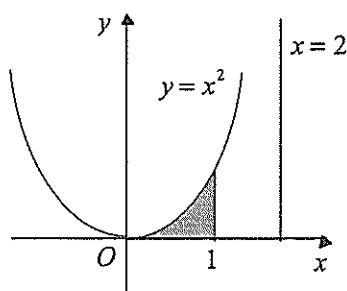
1

- (A) $\frac{1}{2}xe^{\frac{x}{2}} - \frac{1}{4}e^{\frac{x}{2}} + c$
(B) $\frac{1}{2}xe^{\frac{x}{2}} - \frac{1}{2}e^{\frac{x}{2}} + c$
(C) $2xe^{\frac{x}{2}} - 2e^{\frac{x}{2}} + c$
(D) $2xe^{\frac{x}{2}} - 4e^{\frac{x}{2}} + c$

Marks

9

1



The region bounded by the parabola $y = x^2$ and the x -axis between $x = 0$ and $x = 1$ is rotated through one revolution about the line $x = 2$ to form a solid of revolution about the line $x = 2$ to form a solid of volume V . Which of the following is an expression for V ?

- (A) $\pi \int_0^1 (1-x)^2 dy$
- (B) $\pi \int_0^1 (1^2 - x^2) dy$
- (C) $\pi \int_0^1 \{ (2-x)^2 - 1^2 \} dy$
- (D) $\pi \int_0^1 \{ 2^2 - (2-x)^2 \} dy$

10 If $z = \sqrt{3} - i$, then

1

- (A) $z = \sqrt{2} \left(\cos \frac{\pi}{3} - i \sin \frac{\pi}{3} \right)$
- (B) $z = \sqrt{2} \left(\cos \frac{\pi}{6} - i \sin \frac{\pi}{6} \right)$
- (C) $z = 2 \left(\cos \frac{\pi}{3} - i \sin \frac{\pi}{3} \right)$
- (D) $z = 2 \left(\cos \frac{\pi}{6} - i \sin \frac{\pi}{6} \right)$

Section II**90 Marks****Attempt Questions 11 to 16****Allow about 2 hours 45 minutes for this section**

Answer each question in a SEPARATE writing booklet. Extra booklets are available.

All necessary working should be shown in each question.

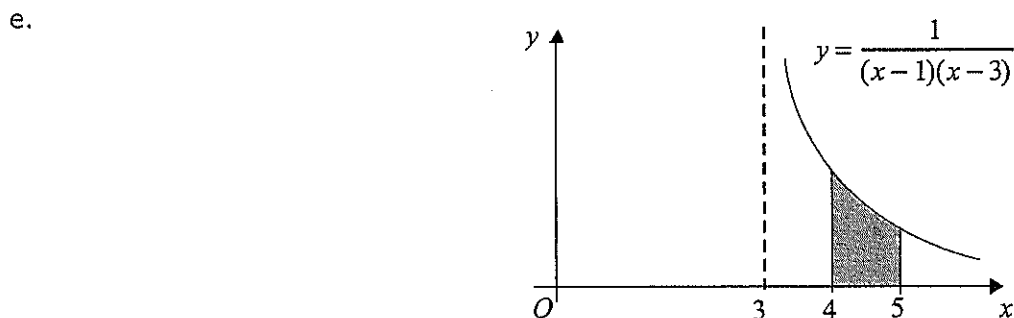
Question 11**Begin a new booklet****Marks**

a. Evaluate $\int_0^{\frac{\pi}{4}} \cos \theta \sin^3 \theta \, d\theta$. **2**

b. Find $\int \frac{\sin 2x + \sin x}{\cos^2 x} \, dx$. **2**

c. Use the substitution $u = 1 + x^2$ to evaluate $\int_0^{\sqrt{3}} \frac{x^3}{(1+x^2)^2} \, dx$ in simplest exact form. **3**

d. Use the substitution $t = \tan \frac{x}{2}$ to evaluate $\int_0^{\frac{\pi}{2}} \frac{1}{2 + \cos x} \, dx$ in simplest exact form. **3**



The region bounded by the curve $y = \frac{1}{(x-1)(x-3)}$ and the x-axis between $x = 4$ and $x = 5$ is rotated through one revolution about the y-axis to form a solid of volume V .

(i) By considering strips of thickness δx perpendicular to the x-axis, use the method of cylindrical shells to show that $V = \pi \int_4^5 \frac{2x}{(x-1)(x-3)} \, dx$. **2**

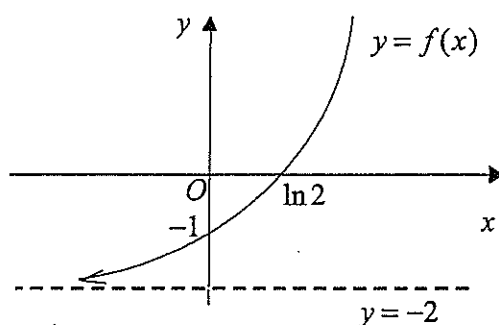
(ii) Hence, find the value of V in simplest exact form. **3**

Question 12**Begin a new booklet****Marks**

- a. Find the values of the real number k such that the equation $x^4 + kx^3 + x^2 + x + 1 = 0$ has an integer root. **2**
- b. Find the complex number $z = a + bi$, where a and b are real, such that $2\bar{z} - iz = 1 + 4i$. **2**
- c. Show that $z = \cos \frac{\pi}{9} + i \sin \frac{\pi}{9}$ is a root of the equation $z^6 - z^3 + 1 = 0$. **2**
- d. In an Argand Diagram, $ABCD$ is a quadrilateral such that vectors $\vec{OA}$, $\vec{OB}$, $\vec{OC}$, $\vec{OD}$ represent the complex numbers a , b , c , d respectively. P , Q , R and S are the midpoints of AB , BC , CD and DA respectively. M and N are the midpoints of PR and QS respectively.
- (i) Show that vectors $\vec{OM}$ and $\vec{ON}$ both represent the complex number $\frac{1}{4}(a + b + c + d)$. **2**
- (ii) Hence explain what type of quadrilateral $PQRS$ is. **1**
- e. The equation $x^4 - kx + 1 = 0$, where k is a real number, has roots α , β , γ and δ .
- (i) Show that $\alpha^2 + \beta^2 + \gamma^2 + \delta^2 = 0$. Hence explain why the equation has either two real and two non-real roots, or four non-real roots. **2**
- (ii) If the equation has a double real root, show that its value is $3^{-\frac{1}{4}}$. **2**
- (iii) Hence show that if the equation has a double real root then each of the two non-real roots has a real part $-3^{-\frac{1}{4}}$ and modulus $3^{-\frac{1}{4}}$. **2**

Question 13**Begin a new booklet****Marks**

- a. The diagram below shows the graph of $f(x) = e^x - 2$.



On separate diagrams sketch the following graphs, in each case showing the intercepts on the axes and the equations of the asymptotes.

- (i) $y = \{f(x)\}^2$. **1**
- (ii) $y = \log_e f(x)$. **1**
- (iii) $y = \frac{1}{f(x)}$. **2**
- (iv) $y^2 = |f(x)|$. **2**
- b. Prove that $\frac{d}{dx} \left[\sqrt{bx - x^2} + \frac{b}{2} \cos^{-1} \left(\frac{2x - b}{b} \right) \right] = -\sqrt{\frac{x}{b - x}}$, for $x \geq 0$. **3**
- c. For $n = 0, 1, 2, 3, \dots$, let $I_n = \int_{e^{-1}}^1 (1 + \log_e x)^n dx$ and $J_n = \int_{e^{-1}}^1 (\log_e x)(1 + \log_e x)^n dx$.
- (i) Show that $I_n = 1 - nI_{n-1}$ for $n = 1, 2, 3, \dots$ **2**
- (ii) Show that $J_n = 1 - (n + 2)I_n$ for $n = 0, 1, 2, 3, \dots$ **2**
- (iii) Hence, find the value of J_3 in simplest exact form. **2**

Question 14**Begin a new booklet****Marks**

- a. The ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, where $a > b > 0$, has eccentricity e .

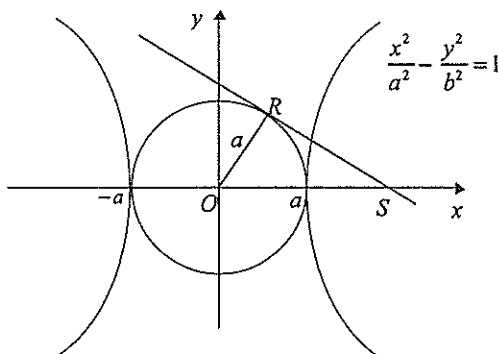
$P(a \cos \theta, b \sin \theta)$ is a point on the ellipse in the first quadrant, S is the focus of the ellipse nearer to P and $Q(a \cos \phi, b \sin \phi)$, $-\pi < \phi \leq \pi$, is a second point on the ellipse so that the normal to the ellipse at Q is parallel to the normal at P .

Point T is the intersection of the tangent at P with the normal at Q .

V lies on the tangent at P so that SV is parallel to QT .

- (i) Show this information on a sketch. **1**
- (ii) Find the gradient of the normal at P by differentiation and deduce that $\phi = \theta - \pi$. **3**
- (iii) Show that the normal at P has x -intercept $ae^2 \cos \theta$. **2**
- (iv) Show that $\frac{VP}{VT} = \frac{1 - e \cos \theta}{1 + e \cos \theta}$ **2**

b.



S is the focus of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, $a \neq b$, which lies on the positive x -axis.

R is a point on the auxiliary circle of the hyperbola such that R lies in the first quadrant and SR is a tangent to the auxiliary circle.

The eccentricity of the hyperbola is e .

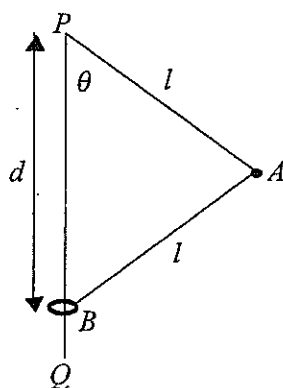
- (i) Show that R lies on a directrix of the hyperbola. **2**
- (ii) Show that SR has equation $y = -\frac{1}{\sqrt{e^2 - 1}}(x - ae)$. **2**
- (iii) If SR meets the hyperbola at the point $(a \sec \theta, b \tan \theta)$, show that **3**
- $$e^2(2 - e^2) \sec^2 \theta - 2e \sec \theta + \{e^2 + (e^2 - 1)^2\} = 0.$$

Question 15**Begin a new booklet****Marks**

- a. A bullet is fired vertically into the air with a speed of 800 ms^{-1} . In the air the bullet experiences air resistance equal to $\frac{mv}{5}$, as well as gravity g . Gravity can be assumed to equal 10 ms^{-2} .

- (i) Find the height reached to the nearest metre. **3**
- (ii) Find the time taken to achieve this height. **2**
- (iii) As the bullet returns to the ground it is subject to the same forces. Find the terminal velocity. **2**

b.



PQ is a smooth vertical rod. Particle A of mass m is attached to point P by a string of length l and A is also attached by a second string of length l to a smooth ring B of mass M which is free to slide on the rod PQ without friction. A is set in motion in a horizontal circle about PQ with angular velocity ω . B is in equilibrium.

- (i) Draw diagrams showing the forces on each of A and B , and hence show that if T_1 and T_2 are the tensions in the strings AP and AB respectively when AP makes an angle θ with the vertical, then $T_1 - T_2 = \frac{mg}{\cos \theta}$, $T_1 + T_2 = ml\omega^2$ and $T_2 = \frac{Mg}{\cos \theta}$. **4**
- (ii) Hence, express the distance d of B below P in terms of g , ω^2 and $\frac{M}{m}$. **3**
- (iii) Deduce that $\omega^2 \geq \frac{g}{l} \left(1 + \frac{2M}{m} \right)$. **1**

Question 16**Begin a new booklet****Marks**

- a. Use Mathematical Induction to show that $3^n - 1 \geq 2n$ for all positive integers $n \geq 1$. **3**
- b. Find the equation of the tangent to the curve $3x^2 - 2xy - y^2 - 20 = 0$ at the point $(3, 1)$. **2**
- c. With respect to the x and y axes, the line $x = 1$ is a directrix and the point $(2, 0)$ is a focus of eccentricity $\sqrt{2}$. Find the equation of the conic, and sketch the curve indicating its asymptotes, foci and directrices. **3**
- d. (i) Prove that $\cot^{-1}(2x - 1) - \cot^{-1}(2x + 1) = \tan^{-1}\left(\frac{1}{2x^2}\right)$, where $x \geq 1$. **3**
- (ii) Find the sum of $K = \tan^{-1}\frac{1}{2} + \tan^{-1}\frac{1}{8} + \tan^{-1}\frac{1}{18} + \dots + \tan^{-1}\left(\frac{1}{2n^2}\right)$, where n is a positive integer. **3**
- (iii) Show that $\lim_{n \rightarrow +\infty} K = \frac{\pi}{4}$. **1**

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

Multiple Choice Answer Sheet

Student Name: _____

Class: _____

1.

A

B

C

D
2.

A

B

C

D
3.

A

B

C

D
4.

A

B

C

D
5.

A

B

C

D
6.

A

B

C

D
7.

A

B

C

D
8.

A

B

C

D
9.

A

B

C

D
10.

A

B

C

D

Section 1

$$\begin{aligned}
 1. \int \frac{\cos^3 x + \sin^3 x}{\cos x + \sin x} dx \\
 &= \int \frac{\cos x + \sin x (\sin^2 x - \sin x \cos x + \cos^2 x)}{\cos x + \sin x} dx \\
 &= \int 1 - \sin x \cos x dx \\
 &= \int 1 - \frac{1}{2} \sin 2x dx \\
 &= x + \frac{1}{4} \cos 2x + C \quad \therefore A
 \end{aligned}$$

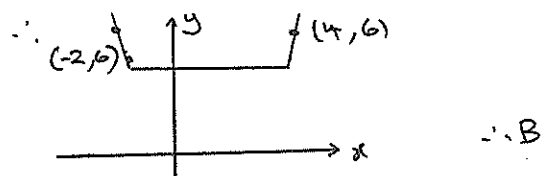
$$\begin{aligned}
 2. \ddot{x} &= \frac{d}{dt} \left[\frac{1}{2} v^2 \right] \\
 &= \frac{d}{dt} \left[\frac{9}{2} (5 + 4x - x^2) \right] \\
 &= \frac{9}{2} (4 - 2x) \\
 &= 9(2 - x) \\
 &= -9(x - 2) \\
 \text{As } \ddot{x} &= -\omega^2 (x - c) \text{ then} \\
 \text{centre of motion at } x &= 2 \quad \therefore C
 \end{aligned}$$

$$\begin{aligned}
 3. \frac{dy}{dx} &= \frac{dy}{d\theta} \cdot \frac{d\theta}{dx} \quad \frac{dy}{d\theta} = \sin \theta \\
 \frac{dx}{d\theta} &= 1 - \cos \theta \\
 \therefore \frac{d\theta}{dx} &= \frac{1}{1 - \cos \theta} \\
 \therefore \frac{dy}{dx} &= \frac{\sin \theta}{1 - \cos \theta} \\
 &= \frac{\sin 2(\theta/2)}{1 - \cos 2(\theta/2)} \\
 &= \frac{2 \sin \theta/2 \cos \theta/2}{1 - [1 - 2 \sin^2 \theta/2]} \\
 &= \frac{2 \sin \theta/2 \cos \theta/2}{2 \sin^2 \theta/2} \\
 &= \cot \theta/2 \quad \therefore B
 \end{aligned}$$

$$\begin{aligned}
 4. |z - (1 - i)| &= 4 \\
 \therefore \text{centre } (1, -1), \text{ radius } 4 \\
 \therefore C
 \end{aligned}$$

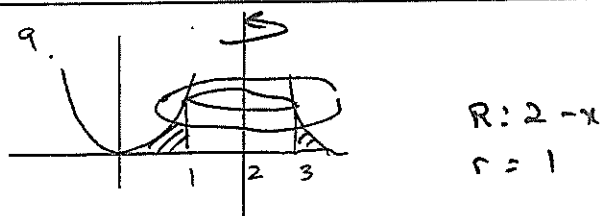
$$\begin{aligned}
 5. \text{If } x = \alpha \text{ is root } \alpha^3 + 2\alpha + 1 &= 0 \\
 \text{Now, if } x = 2\alpha \therefore \alpha &= \frac{x}{2} \text{ is soln.} \\
 \therefore \left(\frac{x}{2}\right)^3 + 2\left(\frac{x}{2}\right) + 1 &= 0 \\
 \therefore \frac{x^3}{8} + x + 1 &= 0 \\
 x^3 + 8x + 8 &= 0 \quad \therefore A
 \end{aligned}$$

$$\begin{aligned}
 6. \text{Subs in } x = -2 \therefore y &= 6 \\
 x = 4 \therefore y &= 6
 \end{aligned}$$



$$\begin{aligned}
 7. \text{Subs } Q \text{ in eqn:} \\
 p^3(c/2) - p\left(\frac{c}{2}\right) &= cp^4 - c \\
 cp^3 \frac{c}{2} - \frac{cp}{2} &= cp^4 - c \\
 p^3 \frac{c}{2} - p &= p^4 \frac{c}{2} - \frac{c}{2} \\
 p^3 \frac{c}{2} \left(\frac{2}{c} - \frac{1}{p}\right) &= \frac{p}{2} \left(\frac{2}{c} - \frac{1}{p}\right) \\
 -p^3 \frac{c}{2} &= 1 \\
 \therefore p^3 \frac{c}{2} &= -1 \quad \therefore B
 \end{aligned}$$

$$\begin{aligned}
 8. \int x e^{x/2} dx \text{ using int by parts:} \\
 u = x \quad v' = e^{x/2} \\
 u' = 1 \quad v = 2e^{x/2} \\
 \therefore 2xe^{x/2} - \int 2e^{x/2} dx \\
 = 2xe^{x/2} - 2 \cdot 2e^{x/2} + C \\
 = 2xe^{x/2} - 4e^{x/2} + C \quad \therefore D
 \end{aligned}$$



$$\begin{aligned}
 \delta v &= \pi(R^2 - r^2) \delta y \\
 \therefore v &= \pi \int_0^1 ((2-y)^2 - 1^2) dy \\
 \therefore C
 \end{aligned}$$

$$10. z = \sqrt{3} - i$$

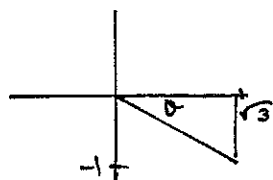
$$|z| = 2$$

$$\arg z = \tan^{-1}\left(\frac{-1}{\sqrt{3}}\right)$$

$$= -\pi/6$$

$$\therefore z = 2(\cos(-\pi/6) + i \sin(-\pi/6))$$

$$= 2(\cos \pi/6 - i \sin \pi/6) \therefore D$$



Question 11

$$a. \int_0^{\pi/4} \cos \theta \sin^3 \theta d\theta$$

$$= \left[\frac{1}{4} \sin^4 \theta \right]_0^{\pi/4}$$

$$= \frac{1}{4} [(\sin \pi/4)^4 - (\sin 0)^4]$$

$$= \frac{1}{4} \left[\left(\frac{1}{\sqrt{2}}\right)^4 - 0 \right]$$

$$= \frac{1}{4} \left[\frac{1}{4} \right]$$

$$= \frac{1}{16}$$

2

$$b. \int \frac{\sin 2x + \sin x}{\cos^2 x} dx$$

$$= \int \frac{2 \sin x \cos x + \sin x}{\cos^2 x} dx$$

$$= \int 2 \frac{\sin x}{\cos x} + \sec x \tan x dx$$

$$= -2 \log(\cos x) + \sec x + C$$

2

$$c. \int_0^{\sqrt{3}} \frac{x^3}{(1+x^2)^{3/2}} dx$$

$$u = 1+x^2$$

$$\frac{du}{dx} = 2x$$

$$dx = \frac{du}{2x}$$

$$x = \sqrt{3} \therefore u = 4$$

$$x = 0 \therefore u = 1$$

$$\int_1^4 \frac{x^2}{(1+x^2)^{3/2}} \cdot \frac{du}{2x}$$

$$= \frac{1}{2} \int_1^4 \frac{u-1}{u^{3/2}} du$$

$$= \frac{1}{2} \int_1^4 u^{-1/2} - u^{-3/2} du$$

$$= \frac{1}{2} \left[2u^{1/2} + 2u^{-1/2} \right]_1^4$$

$$= \left[u^{1/2} + u^{-1/2} \right]_1^4$$

$$= \left(2 + \frac{1}{2} \right) - (1 + 1)$$

3

$$= \frac{1}{2}$$

$$d. \int_0^{\pi/2} \frac{1}{2 + \cos x} dx$$

$$t = \tan \frac{x}{2}$$

$$\frac{dt}{dx} = \frac{1}{2} \sec^2 \frac{x}{2}$$

$$dx = \frac{2 dt}{1+t^2}$$

$$x = \pi/2, t = 1$$

$$x = 0, t = 0$$

$$= \int_0^1 \frac{1}{2 + \frac{1-t^2}{1+t^2}} \cdot \frac{2 dt}{1+t^2}$$

$$= 2 \int_0^1 \frac{1}{2 + 2t^2 + 1 - t^2} dt$$

$$= 2 \int_0^1 \frac{1}{3 + t^2} dt$$

$$= 2 \cdot \frac{1}{\sqrt{3}} \tan^{-1} \frac{t}{\sqrt{3}} \Big|_0^1$$

$$= \frac{2}{\sqrt{3}} \left[\tan^{-1} \frac{1}{\sqrt{3}} - \tan^{-1} 0 \right]$$

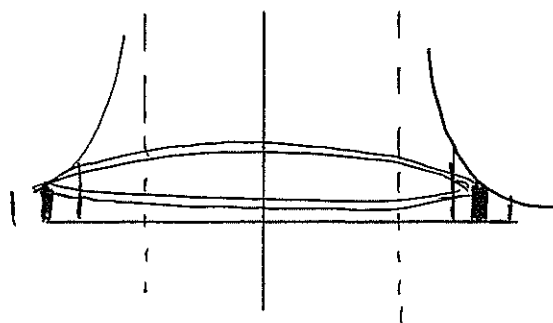
$$= \frac{2}{\sqrt{3}} \cdot \frac{\pi}{6}$$

$$= \frac{\pi}{3\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}}$$

$$= \frac{\pi\sqrt{3}}{9}$$

3

e.



$$\delta V = 2\pi r h \delta x$$

$$\therefore V = 2\pi \int \text{rad. height } dx$$

$$= V = 2\pi \int_4^5 x \cdot \frac{1}{(x-1)(x-3)} dx$$

$$= 2\pi \int_4^5 \frac{x}{(x-1)(x-3)} dx \quad 2$$

$$\text{ii. } \frac{a}{x-1} + \frac{b}{x-3} = \frac{x}{(x-1)(x-3)}$$

$$\therefore a(x-3) + b(x-1) = x$$

$$x=3 \therefore 2b=3 \therefore b=\frac{3}{2}$$

$$x=1 \therefore -2a=1 \therefore a=-\frac{1}{2}$$

$$\therefore 2\pi \int_4^5 \frac{-\frac{1}{2}}{x-1} + \frac{\frac{3}{2}}{x-3} dx$$

$$= \pi \int_4^5 \frac{-1}{x-1} + \frac{3}{x-3} dx$$

$$= \pi [3 \ln(x-3) - \ln(x-1)]_4^5$$

$$= \pi [3 \ln 2 - \ln 4 - (3 \ln 1 - \ln 3)]$$

$$= \pi [3 \ln 2 - \ln 4 + \ln 3]$$

$$= \pi [\ln 8 - \ln 4 + \ln 3]$$

$$= \pi [\ln 2 + \ln 3]$$

$$= \pi \ln 6 \quad 3$$

Question 12

a. As constant term is 1 then the root (α) is a factor of 1

ie two cases:

$$\alpha=1 \therefore x=1 : 1+k+1+1+1=0$$

$$\therefore k=-4$$

$$\alpha=-1 \therefore x=-1 : 1-k+1-1+1=0$$

$$\therefore k=2$$

$$\therefore k=2, -4 \quad 2$$

b. $z=a+bi \therefore \bar{z}=a-bi$

$$\therefore 2(a-bi) - i(a+bi) = 1+4i$$

$$\therefore 2a+b=1$$

$$-2b-a=4$$

$$\text{ie } 2a+b=1 \quad \text{--- (1)}$$

$$a+2b=-4 \quad \text{--- (2)}$$

$$2 \times (1) \quad 4a+2b=2 \quad \text{--- (3)}$$

$$(3) - (2) \quad 3a=6$$

$$a=2$$

Substn (1) $4+b=1$

$$b=-3$$

$$\therefore 2-3i \quad 2$$

c. $z = \cos \theta + i \sin \theta$

$$z^3 = \cos 3\theta + i \sin 3\theta$$

$$= \frac{1}{2} + i \frac{\sqrt{3}}{2}$$

$$z^6 = \cos 2\theta + i \sin 2\theta$$

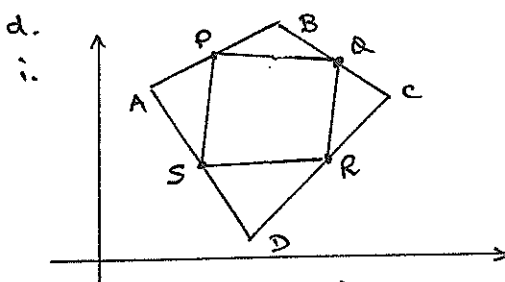
$$= -\frac{1}{2} + i \frac{\sqrt{3}}{2}$$

$$\therefore z^6 - z^3 + 1 = -\frac{1}{2} + i \frac{\sqrt{3}}{2} - \left(\frac{1}{2} + i \frac{\sqrt{3}}{2} \right) + 1$$

$$= -1 + 0 + 1$$

$$= 0$$

$$\therefore z^6 - z^3 + 1 = 0 \quad 2$$



$$\vec{OP} = \vec{OA} + \vec{AP}$$

$$\therefore \vec{OP} = \vec{OA} + \frac{1}{2} \vec{AB}$$

$$\therefore \vec{OP} = a + \frac{1}{2}(b-a)$$

$$= a + \frac{1}{2}b - \frac{1}{2}a$$

$$= \frac{1}{2}(a+b)$$

$$\text{Similarly } \vec{OR} = \frac{1}{2}(c+d).$$

Now, M is midpoint of PR,

$$\therefore \vec{OM} = \frac{1}{2} \left[\frac{1}{2}(a+b) + \frac{1}{2}(c+d) \right]$$

$$= \frac{1}{4}(a+b+c+d)$$

Similarly, since N is midpoint of QS,

$$\text{then } \vec{ON} = \frac{1}{2} \left[\frac{1}{2}(b+c) + \frac{1}{2}(d+a) \right]$$

$$= \frac{1}{4}(a+b+c+d) \quad 2$$

ii. " and N are the same point.
 $\therefore$ diagonals of quadrilaterals bisect each other $\therefore$ parallelogram

e. i. $x^4 - kx + 1 = 0$.

$\therefore \alpha + \beta + \gamma + \delta = -\frac{b}{a} = 0$

$\alpha\beta + \alpha\gamma + \alpha\delta + \beta\gamma + \beta\delta + \gamma\delta = \frac{c}{a} = 0$

Now, $\alpha^2 + \beta^2 + \gamma^2 + \delta^2$

$= (\alpha + \beta + \gamma + \delta)^2 - 2[\alpha\beta + \alpha\gamma + \alpha\delta + \beta\gamma + \beta\delta + \gamma\delta]$

$= 0$

Now, 0 is not root and roots cannot all be real if Σ of squares is 0. As coefficients real, then roots appear in conjugate pairs. This means 2 non-real or 4 non-real.

ii. Let roots be $\alpha, \alpha, \beta, \bar{\beta}$

$P(x) = x^4 - kx + 1$

$P'(x) = 4x^3 - k$

$\therefore \alpha^4 - k\alpha + 1 = 0$ — (1)

$4\alpha^3 - k = 0$ — (2)

$\alpha \times (2) \quad 4\alpha^4 - k\alpha = 0$ — (3)

(3) - (1) $3\alpha^4 - 1 = 0$

$\alpha^4 = \frac{1}{3}$

$\alpha = 3^{-\frac{1}{4}}$

2

iii. $2\alpha + \beta + \bar{\beta} = 0$

$\therefore 2\operatorname{Re}(\beta) = -2\alpha$

$\operatorname{Re}(\beta) = -\alpha$

$\therefore \operatorname{Re}(\beta) = -3^{-\frac{1}{4}}$

Hence, $\operatorname{Re}(\bar{\beta}) = -3^{-\frac{1}{4}}$

Also, $\alpha \cdot \alpha \cdot \beta \cdot \bar{\beta} = 1$

$\therefore |\beta|^2 = \frac{1}{\alpha^2}$

$|\beta|^2 = 3^{\frac{1}{2}}$

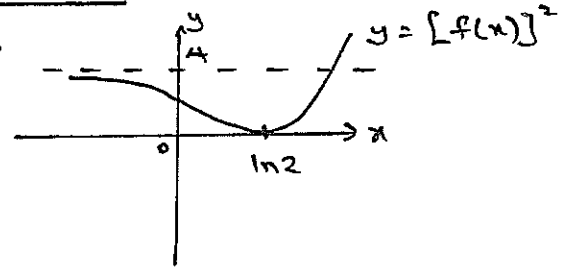
$\therefore |\beta| = 3^{\frac{1}{4}}$

Also, $|\bar{\beta}| = 3^{\frac{1}{4}}$

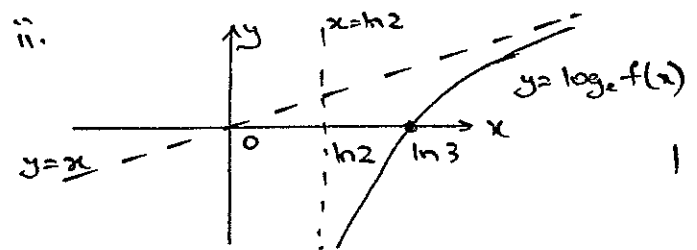
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Question 12

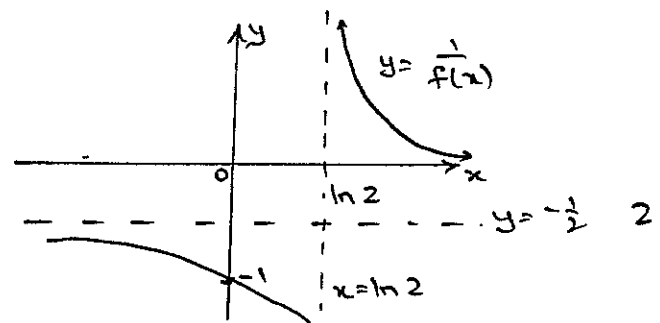
a i.



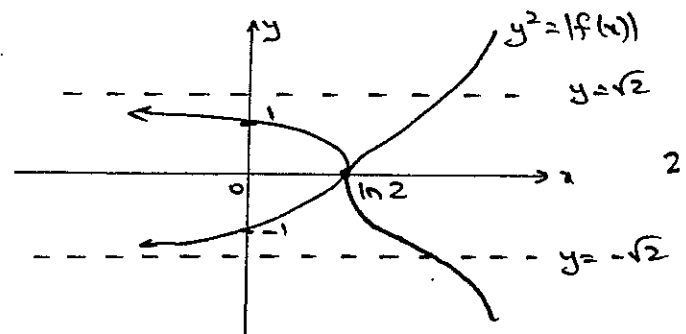
ii.



iii.



iv.



b. $\frac{d}{dx} \left[(bx - x^2)^{\frac{1}{2}} + \frac{b}{2} \cos^{-1} \frac{2x-b}{b} \right]$
 $= \frac{1}{2}(bx - x^2)^{-\frac{1}{2}} (b - 2x) + \frac{b}{2} \frac{-\frac{2}{b}}{\sqrt{1 - \left(\frac{2x-b}{b}\right)^2}}$
 $= \frac{b-2x}{2\sqrt{bx-x^2}} - \frac{1}{\sqrt{\frac{b^2 - (2x-b)^2}{b^2}}}$
 $= \frac{b-2x}{2\sqrt{bx-x^2}} - \frac{b}{\sqrt{b^2 - 4x^2 + 4xb - b^2}}$
 $= \frac{b-2x}{2\sqrt{bx-x^2}} - \frac{b}{\sqrt{4bx-4x^2}}$
 $= \frac{b-2x}{2\sqrt{bx-x^2}} - \frac{b}{2\sqrt{bx-x^2}}$
 $= \frac{-x}{\sqrt{bx-x^2}}$

$$= -\frac{x^2}{\sqrt{bx-x^2}} \quad \text{for } x \geq 0$$

$$= -\sqrt{\frac{x}{b-x}} \quad 3$$

c.i. Let $(1+\log_e x)^n = (1+\log_e u)^n \cdot 1$
 $\quad \quad \quad \uparrow \quad \quad \quad \uparrow$
 $\quad \quad \quad u \quad \quad \quad v'$

$$\therefore I_n = \int_{e^{-1}}^1 (1+\log_e x)^n$$

$$= x(1+\log_e x)^n \Big|_{e^{-1}}^1 - \int_{e^{-1}}^1 x \cdot n(1+\log_e x)^{n-1} \cdot \frac{1}{x} dx$$

$$= 1 - \frac{1}{e} (1+\log_e \frac{1}{e})^n - \int_{e^{-1}}^1 n \cdot (1+\log_e x)^{n-1} dx$$

$$= 1 - n I_{n-1} \quad \text{for } n=1, 2, 3, \dots \quad 2$$

ii. Let $(\log_e x)(1+\log_e x)^n$

$$= (1+\log_e x - 1)(1+\log_e x)^n$$

$$= (1+\log_e x)^{n+1} - (1+\log_e x)^n$$

$$\therefore J_n = I_{n+1} - I_n$$

$$= [1 - (n+1)I_n] - I_n$$

$$= 1 - (n+2)I_n \quad \text{for } n=0, 1, 2, \dots \quad 2$$

iii. $J_3 = 1 - 5I_3$

$$= 1 - 5(1 - 3I_2)$$

$$= -4 + 15I_2$$

$$= -4 + 15(1 - 2I_1)$$

$$= 11 - 30I_1$$

$$= 11 - 30(1 - I_0)$$

$$= -19 + 30I_0$$

$$\text{Now } I_0 = \int_{e^{-1}}^1 1 dx$$

$$= x \Big|_{e^{-1}}^1$$

$$= 1 - \frac{1}{e}$$

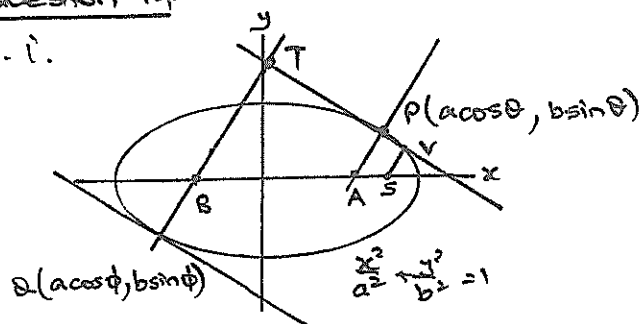
$$\therefore J_3 = -19 + 30 \left[1 - \frac{1}{e} \right]$$

$$= 11 - \frac{30}{e} \quad 2$$

Question 14

5

a. i.



ii. $x = a \cos \theta \quad \frac{dx}{d\theta} = -a \sin \theta$

$$y = b \sin \theta \quad \frac{dy}{d\theta} = b \cos \theta$$

$$\frac{dy}{dx} = \frac{dy}{d\theta} \cdot \frac{d\theta}{dx}$$

$$= b \cos \theta \cdot \frac{1}{-a \sin \theta}$$

$$= \frac{b \cos \theta}{-a \sin \theta}$$

$\therefore$ grad of normal: $\frac{a \sin \theta}{b \cos \theta}$

Now, grads of normals at P, Q equal:

$$\therefore \frac{a \sin \theta}{b \cos \theta} = \frac{a \sin \phi}{b \cos \phi}$$

$$\therefore \tan \theta = \tan \phi$$

Now, $\theta = \phi$, or $\theta = \pi + \phi$

But $\theta \neq \phi \therefore \theta = \pi + \phi$
 ie $\phi = \theta - \pi \quad 3$

iii. Normal at P:

$$y - b \sin \theta = \frac{a \sin \theta}{b \cos \theta} [x - a \cos \theta]$$

Let $y=0 \therefore -b^2 \sin \theta \cos \theta = a \sin \theta \cdot x - a^2 \sin \theta \cos \theta$

$$\therefore ax = \frac{\cancel{\sin \theta} \cos \theta (a^2 - b^2)}{\cancel{\sin \theta}}$$

$$\therefore ax = \cos \theta (a^2 - b^2)$$

Now, $b^2 = a^2(1 - e^2)$

$$\text{ie } b^2 = a^2 - a^2 e^2$$

$$a^2 e^2 = a^2 - b^2$$

$$\therefore ax = \cos \theta \cdot a^2 e^2$$

$$\therefore x = ae^2 \cos \theta \quad 2$$

iv. Similarly, x-int where normal at Q meets x axis: $-ae^2 \cos \theta$

Let normals at P & Q meet at A and B respectively.

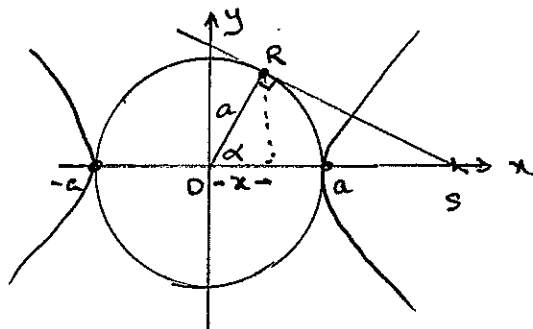
$$\therefore \frac{VP}{VT} = \frac{SA}{SB} \quad (\text{ratio of intercepts in prop}^n)$$

$$\begin{aligned} \therefore \frac{VP}{VT} &= \frac{ae - ae^2 \cos \theta}{ae + ae^2 \cos \theta} \\ &= \frac{ae(1 - e \cos \theta)}{ae(1 + e \cos \theta)} \\ &= \frac{1 - e \cos \theta}{1 + e \cos \theta} \end{aligned}$$

2

b.

i.



Let $\angle ROS = \alpha$

$$\therefore \sec \alpha = \frac{OS}{OR} = \frac{ae}{a} = e \quad \therefore \sec \alpha = e$$

$$\therefore \cos \alpha = \frac{1}{e}$$

Now, at R, $\frac{x}{a} = \cos \alpha$

$$\therefore x = a \cos \alpha$$

$$\therefore x = \frac{a}{e}$$

But directrix is $x = \frac{a}{e}$

$\therefore$ on directrix

2

ii. Subs $x = \frac{a}{e}$ in $x^2 + y^2 = a^2$

$$\therefore \frac{a^2}{e^2} + y^2 = a^2$$

$$y^2 = a^2 - \frac{a^2}{e^2}$$

$$= \frac{a^2 e^2 - a^2}{e^2}$$

$$= \frac{a^2 (e^2 - 1)}{e^2}$$

$$\therefore y = \frac{a}{e} \sqrt{e^2 - 1}$$

$$\therefore \text{grad of OR} = \frac{\frac{a}{e} \sqrt{e^2 - 1}}{e} = \frac{a}{e^2}$$

$$= \sqrt{e^2 - 1}$$

$$\therefore \text{grad of SR} = \frac{-1}{\sqrt{e^2 - 1}}$$

$$\therefore \text{eqn of SR: } y - 0 = \frac{-1}{\sqrt{e^2 - 1}} (x - ae) \quad \text{--- (1)}$$

iii. Subs $(a \sec \theta, b \tan \theta)$ in (1)

$$b \tan \theta (\sqrt{e^2 - 1}) = -(a \sec \theta - ae)$$

Square both sides

$$b^2 \tan^2 \theta (e^2 - 1) = a^2 (\sec \theta - e)^2$$

$$\frac{b^2}{a^2} [(\sec^2 \theta - 1)(e^2 - 1)] = (\sec \theta - e)^2$$

$$\text{But } \frac{b^2}{a^2} (e^2 - 1)$$

$$\therefore (e^2 - 1)^2 (\sec^2 \theta - 1)$$

$$= \sec^2 \theta - 2e \sec \theta + e^2$$

$$(e^4 - 2e^2 + 1)(\sec^2 \theta - 1)$$

$$= \sec^2 \theta - 2e \sec \theta + e^2$$

$$(e^4 - 2e^2)(\sec^2 \theta - 1) = -(\sec^2 \theta - 1) + \sec^2 \theta - 2e \sec \theta + e^2$$

$$(e^4 - 2e^2) \sec^2 \theta = 1 - 2e \sec \theta + e^2 + e^4 - 2e^2$$

$$e^2 (e^2 - 2) \sec^2 \theta = -2e \sec \theta + e^4 - e^2 +$$

$$e^2 (2 - e^2) \sec^2 \theta - 2e \sec \theta$$

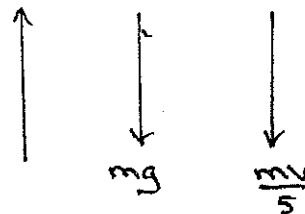
$$+ e^4 - 2e^2 + 1 + e^2 = 0$$

$$\therefore e^2 (2 - e^2) \sec^2 \theta - 2e \sec \theta$$

$$+ [e^2 + (e^2 - 1)^2] = 0 \quad \text{--- (3)}$$

Question 15

a. i.



$$\therefore m\ddot{x} = -\frac{m\dot{v}}{s} - mg$$

$$\text{ie } \ddot{x} = -\frac{\dot{v}}{s} - g$$

To find height, use $\ddot{x} = v \frac{dv}{dh}$

$$\therefore v \frac{dv}{dt} = -\frac{v}{5} - g$$

$$\frac{dv}{dt} = -\frac{v+5g}{5v}$$

$$\therefore \frac{dt}{dv} = \frac{-5v}{v+5g}$$

$$\therefore t = -5 \int_{800}^0 \frac{v}{v+5g} dv = \int_0^{800} 5 \frac{v}{v+5g} dv$$

$$= 5 \int_0^{800} \frac{v+5g-5g}{v+5g} dv$$

$$= 5 \int_0^{800} 1 - \frac{5g}{v+5g} dv$$

$$= 5 \left[v - 5g \ln(v+5g) \right]_0^{800}$$

$$\text{Let } g=10$$

$$\therefore t = 5 \left[800 - 50 [\ln 850 - \ln 50] \right]$$

$$= 5 \left[800 + 50 \ln \left(\frac{850}{50} \right) \right]$$

$$\approx 3292 \text{ m}$$

$$\text{ii. For time, let } \ddot{x} = \frac{dv}{dt}$$

$$\therefore \frac{dv}{dt} = -\frac{v}{5} - g$$

$$= -\frac{v+5g}{5}$$

$$\therefore \frac{dt}{dv} = \frac{-5}{v+5g}$$

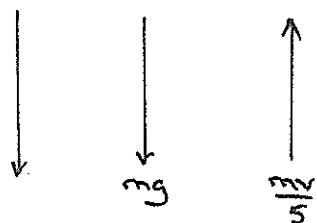
$$t = -5 \int_{800}^0 \frac{1}{v+5g} dv = 5 \int_0^{800} \frac{1}{v+5g} dv$$

$$= -5 \ln(v+5g) \Big|_0^{800}$$

$$= -5 (\ln 850 - \ln 50)$$

$$\approx 14.17 \text{ s}$$

iii.



$$\therefore m\ddot{x} = mg - \frac{mv}{5}$$

$$\therefore \ddot{x} = g - \frac{v}{5}$$

$$\therefore \ddot{x} = \frac{5g-v}{5}$$

For terminal velocity,

$$\ddot{x} = 0$$

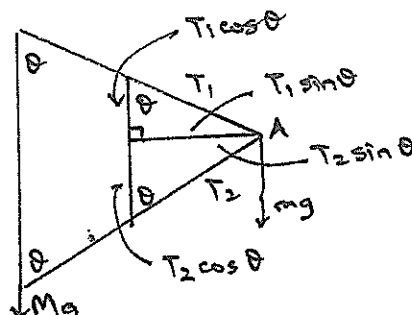
$$\therefore 5g-v=0$$

$$\therefore v=5g$$

$$\text{As } g=10 \therefore v=50 \text{ ms}^{-1}$$

b.

i.



Forces on A:

$$\text{Vertically: } T_1 \cos \theta - T_2 \cos \theta = mg$$

$$\therefore (T_1 - T_2) \cos \theta = mg$$

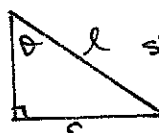
$$\therefore T_1 - T_2 = \frac{mg}{\cos \theta} \quad \text{--- (1)}$$

$$\text{Horizontally: } T_1 \sin \theta + T_2 \sin \theta = m r \omega^2$$

$$\therefore (T_1 + T_2) \sin \theta = m r \omega^2$$

$$\therefore T_1 + T_2 = \frac{m r \omega^2}{\sin \theta}$$

But



$$\sin \theta = \frac{r}{l} \therefore l = \frac{r}{\sin \theta}$$

$$\therefore T_1 + T_2 = m l \omega^2 \quad \text{--- (2)}$$

Forces on B:

$$\text{Vertically: } T_2 \cos \theta = Mg$$

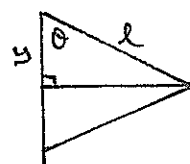
$$\therefore T_2 = \frac{Mg}{\cos \theta} \quad \text{--- (3)}$$

$$\text{ii. (2) - (1) } 2T_2 = m l \omega^2 - \frac{mg}{\cos \theta} \quad \text{--- (4)}$$

$$\text{Subs (4) in (3) } \frac{2Mg}{\cos \theta} = m l \omega^2 - \frac{mg}{\cos \theta}$$

$$\therefore 2Mg = m l \omega^2 \cos \theta - mg$$

Now, as:



$$y = l \cos \theta$$

$$\therefore d = 2l \cos \theta$$

$$\therefore \frac{d}{2} = l \cos \theta$$

$$\therefore 2Mg = \frac{m d \omega^2}{2} - mg$$

$$4Mg = m d \omega^2 - 2mg$$

$$mdw^2 = 2g[2M+m]$$

$$d = \frac{2g}{mw^2}[2M+m]$$

$$\therefore d = \frac{2g}{w^2} \left[1 + \frac{2M}{m} \right] \quad 3$$

iii. Now, $d \leq 2\lambda$

$$\therefore \frac{2g}{w^2} \left[1 + \frac{2M}{m} \right] \leq 2\lambda$$

$$\therefore w^2 \geq \frac{2g}{2\lambda} \left[1 + \frac{2M}{m} \right]$$

$$\therefore w^2 \geq \frac{g}{\lambda} \left[1 + \frac{2M}{m} \right] \quad 1$$

Question 16:

a. Step 1: Prove true for $n=1$

$$\therefore 3^1 - 1 \geq 2(1)$$

ie $2 \geq 2$ Yes!

$\therefore$ true for $n=1$

Step 2: Assume true for $n=k$

$$\therefore 3^k - 1 \geq 2k$$

Now prove true for $n=k+1$

$$\therefore 3^{k+1} - 1 \geq 2(k+1)$$

$$\text{Now, } 3^{k+1} - 1 = 3 \cdot 3^k - 1$$

$$\geq 3 \cdot (2k+1) - 1$$

$$= 3 \cdot 2k + 2$$

$$= 2k + 2 + 4$$

$$= 2(k+1) + 4$$

$$\geq 2(k+1)$$

$$\therefore 3^{k+1} - 1 \geq 2(k+1)$$

$\therefore$ true for $n=k+1$

Step 3: As true for $n=1$, then true for $n=2, 3, \dots$ for all true integers. 3

$$b. 6x - 2y - 2x \frac{dy}{dx} - 2y \frac{dy}{dx} = 0$$

Subs (3,1):

$$\therefore 18 - 2 - 6 \frac{dy}{dx} - 2 \frac{dy}{dx} = 0$$

$$\therefore 8 \frac{dy}{dx} = 16$$

$$\therefore \frac{dy}{dx} = 2$$

$$\therefore y - 1 = 2(x - 3)$$

$$y - 1 = 2x - 6$$

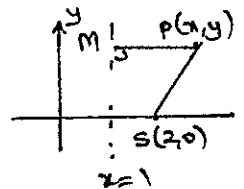
$$\therefore 2x - y - 5 = 0 \quad 2$$

c. Let $P(x, y)$ be a point on conic and let $S(2, 0)$. M is foot of $\perp$ from P to directrix.

$$\therefore \frac{PS}{PM} = e$$

$$\frac{PS^2}{PM^2} = 2$$

$$PS^2 = 2 \times PM^2$$

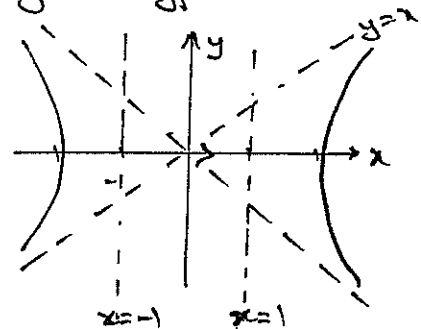


$$\therefore (x-2)^2 + y^2 = 2(x-1)^2$$

$$x^2 - 4x + 4 + y^2 = 2x^2 - 4x + 2$$

$$\therefore x^2 - y^2 = 2$$

Eqn of hyperbola, with asymptotes $y = \pm x$. As asymptotes are $\perp$ then it is rectangular hyperbola.



$$d. i. \text{ Let } \alpha = \cot^{-1}(2x-1)$$

$$\cot \alpha = 2x-1$$

$$\text{Let } \beta = \cot^{-1}(2x+1)$$

$$\cot \beta = 2x+1$$

$$\text{Now, } \cot(\cot^{-1}(2x-1) - \cot^{-1}(2x+1))$$

$$= \cot(\alpha - \beta)$$

$$= \frac{1}{\tan(\alpha - \beta)}$$

$$= \frac{1 + \tan \alpha \tan \beta}{\tan \alpha - \tan \beta}$$

$$= \frac{1 + \frac{1}{\cot \alpha} \cdot \frac{1}{\cot \beta}}{\frac{1}{\cot \alpha} - \frac{1}{\cot \beta}}$$

$$\frac{1}{\cot \alpha} - \frac{1}{\cot \beta}$$

$$= \frac{\cot \alpha \cot \beta + 1}{\cot \beta - \cot \alpha}$$

$$= \frac{(2x-1)(2x+1) + 1}{2x-1 - (2x+1)}$$

$$= \frac{4x^2 - 1 + 1}{2}$$

$$= 2x^2$$

$$\therefore \tan(\alpha - \beta) = \frac{1}{2x^2}$$

$$\therefore \alpha - \beta = \tan^{-1} \frac{1}{2x^2} \quad 3$$

$$\therefore \cot^{-1}(2x-1) - \cot^{-1}(2x+1) = \tan^{-1}\left(\frac{1}{2x^2}\right)$$

$$\text{ii. Let } x=1 \therefore \tan^{-1} \frac{1}{2} = \cot^{-1} 1 - \cot^{-1} 3$$

$$\text{Let } x=2 \therefore \tan^{-1} \frac{1}{8} = \cot^{-1} 3 - \cot^{-1} 5$$

$$\text{Let } x=n \therefore \tan^{-1} \frac{1}{2n^2} = \cot^{-1}(2n-1) - \cot^{-1}(2n+1)$$

By adding equations,

$$\tan^{-1} \frac{1}{2} + \tan^{-1} \frac{1}{8} + \dots + \tan^{-1} \frac{1}{2n^2}$$

$$= \cot^{-1} 1 - \cancel{\cot^{-1} 3} + \cancel{\cot^{-1} 3} - \cancel{\cot^{-1} 5} + \dots + \dots - \cot^{-1}(2n+1)$$

$$= \overset{\textcircled{4}}{\cancel{\cot^{-1} 1}} - \cot^{-1}(2n+1) \quad 3$$

$$\therefore K = \textcircled{4} - \cot^{-1}(2n+1)$$

$$\text{iii. } \lim_{n \rightarrow \infty} \textcircled{4} - \cot^{-1}(2n+1)$$

$$n \rightarrow \infty$$

$$\text{As } n \rightarrow \infty, 2n+1 \rightarrow \infty, \cot^{-1} \rightarrow 0$$

$$\therefore \lim_{n \rightarrow \infty} \textcircled{4} - \cancel{\cot^{-1}(2n+1)}_0 = \textcircled{4}$$